



granted.bio: Beta Review of a Research Intelligence Report

Report ID 293caa10 · "Cell-free Antibody Engineering" · 53 pages, generated 4 August 2026.

Prepared for Ted Nunes, i5 BioPartners From Nathan Gardner, PhD 11 August 2026

Reviewed cover to cover, from the perspective the report is least likely to have been tested against: someone who has spent a career inside cell-free protein expression. Findings are ordered by what would change the product most, and every one carries a page number so you can go straight to it.

1 What already works, and is worth protecting

- **"What This Report Does Not Cover" (p.2 to 3) is genuinely rare.** NIH-only, US-only, gated on grant acknowledgment, RePORTER lag, umbrella-grant exclusion, automated categorization: disclosed *before* the findings rather than buried in a methods note. Most competitive-intelligence products hide their sampling frame. Do not let this get trimmed.
- **Excluding umbrella P30 and CTSA dollars from Total Committed Funding** (p.30, p.52) to keep the headline number topically attributable is a thoughtful, non-obvious call, and you say so.
- **Confidence plus Evidence on claims** is the right instinct, and "What Surprised Us" framed as "*flagged hypotheses, not verified conclusions*" is the correct posture for algorithmic findings.
- **Market Context (p.37 to 41) is the strongest section in the report.** Sutro's XpressCF scale-up, Nuclera's Series C and Antibody Triage Service, LenioBio's ALiCE HTPE, the CFPS market sizing: accurate, current, and commercially useful. I would read this section on any topic.

2 The headline finding: the sample does not contain the topic

The query was "cell-free antibody engineering using in vitro transcription translation and cell-free protein synthesis systems" (p.52). The returned sample contains essentially no cell-free protein synthesis. This is not my inference. It is in your own tables.

Production & Expression Platform, p.15. Cell-Free Expression: **0 projects, 0.0%, \$0.0M.** Phage Display 0. Yeast 0. HEK293 0. Ribosome/mRNA Display 1.

The largest category in that same table is **B Cell & Hybridoma at 28 projects and \$42.9M**, and the largest translational stage is In Vivo Preclinical at 22 projects and \$35.5M. In vivo B cell engineering is not adjacent to cell-free expression. It is the opposite of it.

- **None of the five methodological clusters (p.8 to 10) is cell-free protein synthesis:** in vivo and ex vivo B cell engineering, machine-learning design, continuous evolution, glycoengineering, mRNA delivery.
- **None of the ten top-funded projects (p.32 to 37) uses a cell-free expression system.**
- **The report already knows this and says it on p.17:** "Cell-Free Expression, the production approach most directly named in the topic, registers 0 projects in the sample." That single sentence is the most important thing in 53 pages, and it is on page 17, inside a section a reader is likely to skim.

The "Precise" signal is being read backwards

p.53 reports all 118 matches as Precise (similarity 50% or above) and interprets that as "*the topic mapping to a tightly-bounded research area.*" Read against the p.15 table, it means the opposite. A 100% Precise rate combined with zero projects

in the category named by the query's own head term is the signature of a query **collapsing onto a broader concept**, in this case plain "antibody engineering". The embedding treated "cell-free" as a weak modifier and dropped it. High similarity to a query that lost half its meaning is not precision.

The single highest-value change I would make. Turn this into an automatic pre-flight check. When a white-space category that matches the query's own head term returns zero projects, warn before generating 53 pages: "*Your sample contains no projects using cell-free expression. Consider narrowing the query or proceeding with a scope caveat.*" That check is low-cost to build, it is computable from data you already have on the page, and it converts your most damaging failure mode into a moment where the product looks smarter than the user.

One diagnostic worth sitting with: the section you label "supplementary" (Market Context, web-sourced) was the only part of this report actually about cell-free antibody engineering. The core NIH semantic-search pipeline missed the topic; the live web search hit it.

3 When the sample misses, the insight generator invents the connection

This is the pattern I would worry about most in a paid deliverable. Every one of the ten top-funded project entries carries a sentence explaining its cell-free relevance, and because the underlying grants have none, those sentences are manufactured. Four of them are wrong on the science, in ways a domain reader will catch immediately.

Page	What the report says	Why it does not hold
p.33	Plasma cell project: informs " <i>which cell-free reconstitution components, transcription factors, cytokine mimetics, might be necessary to replicate high-level antibody secretion.</i> "	Transcription factors and cytokine mimetics are not components of a cell-free reaction. You do not reconstitute plasma-cell secretory identity in a lysate, and a lysate has no secretory pathway to begin with.
p.33	Bone marrow organoid " <i>sits conceptually between fully cell-based and fully cell-free approaches.</i> "	An organoid is emphatically cell-based. There is no axis on which it sits between the two.
p.34	Adenoviral vector payloads " <i>require iterative cell-free expression testing to confirm construct functionality before packaging.</i> "	Not required, and not standard. Construct validation here is normally done in cells. Cell-free would be a reasonable <i>suggestion</i> ; stating it as a requirement is a claim about the field that is not true.
p.32	The #1 top-funded project at \$7.2M, a structural and computational core, gets three sentences of cell-free relevance.	The entry concedes in the same paragraph that there is no public abstract . Worth a data spot-check too: the PI is listed at Washington University, while Richard J. Kuhn is a flavivirus structural biologist at Purdue. However that resolves, a structural virology core is not a cell-free antibody engineering project.

The same pattern reaches the Must-Read Publications (p.46 to 47)

Three of the four have no cell-free content at all, and each is given a cell-free framing that the paper does not support. I checked all four against PubMed:

- **In vivo affinity maturation of reprogrammed mouse B cells** (Nat Biomed Eng 2024). Mouse B cells, CRISPR, in vivo. No cell-free content. The report says it "*bridges cell-free antibody engineering with in vivo optimization.*"
- **Nanovial plasma B cell screening** (bioRxiv 2024). Entirely cell-based, an industry preprint from Alloy Therapeutics and Partillion. The report frames it as yielding antibodies "*without relying on traditional cell-free display alone.*"

- **The Residue Fidelity Index paper** (BioTechniques). This one I would pull. It is **single-author, from a commercial vendor** (Precision BioTools), validated only on a **simulated testbed with proprietary immunogens**, and it proposes the author's own branded modules. The report also dates it **2025; PubMed has it at 29 January 2026**. It is a vendor's metric proposal, not must-read literature for this field.

The fix is a rule, not a better prompt. When a project or paper has no demonstrable link to the query's core method, say so rather than generating a bridge. An honest *"no direct cell-free link; included for landscape context"* costs you nothing and buys the reader's trust in every other sentence. Right now a domain expert who spots one invented bridge starts discounting the whole report, including the parts that are correct.

4 Accuracy defects, ranked by damage per unit of effort to fix

Page	Defect	Fix
CRITICAL: VISIBLE TO ANY READER, UNDERMINES THE WHOLE DOCUMENT		
p.28	Raw model self-correction shipped in the prose, twice: "...5R33CA272331-03 is Minnesota - correcting: UC Irvine projects include...", then the identical string again in the Boston University paragraph.	A regex or validation pass for self-referential editing language before render. This is the highest damage-to-effort ratio item in the review: it tells a paying reader the text was not read by anyone before it shipped.
HIGH: FACTUAL ERRORS A TARGET READER WILL CATCH		
p.40	"Thermo Fisher's PURExpress kit" . PURExpress is New England Biolabs. Your own Key Players list states this correctly on the previous page, so the report contradicts itself across two pages.	Validate vendor and product pairings against a controlled list. This is a first-week fact in cell-free; getting it wrong is disproportionately costly with this audience.
p.39	"Merck KGaA (ReadyToProcess CFPS line)" . ReadyToProcess is Cytiva, now Danaher, and it is single-use bioprocess consumables and columns, not a cell-free line at all. Merck's bioprocess brand is Mobius.	Wrong company and wrong product category in one entry. Same controlled-list fix.
p.16 vs p.17	A percentage printed as dollars. The table gives In Vivo B Cell Engineering as 4 projects, 3.4%, \$7.9M . The prose two paragraphs later says "(4 projects, \$3.4M)". It read the share column into the funding slot.	A genuine template bug, and the kind that erodes trust in every other number. Worth grepping for across all generated sections, since the same helper probably feeds them.
COSMETIC, BUT THEY READ AS CARELESSNESS		
p.49	Duplicate PI row: "MILLER, SHANNON MARIE" and "Shannon Marie Miller" listed separately, same \$3.4M, same organization.	Case-insensitive name normalization. It also means the headline "130 Principal Investigators" is inflated by an unknown amount.
p.40, 48, 50	Stray characters: Boston University (Charles River Campus)), HOWELL, DAVID W. (contact)), and Cyrillic glyphs inside REFRAme-01.	Unbalanced parenthesis on name suffixes, plus a Unicode normalization pass on web-scraped strings.

5 Length: 53 pages is roughly an hour, and there is no table of contents

About 15,150 words. My honest reaction on receiving it was that I was not going to read it all, and I am the ideal reader for this topic. The problem is not that the analysis is too deep. It is that reference material and reading material are interleaved.

What repeats	Count	Comment
"Keywords per category (what was counted)" blocks	5 blocks, ~1,550 words	Roughly 5 pages of raw keyword lists. This is QA material, not reading material.
The dagger footnote, verbatim	5×	Identical text each time. Define once, reference thereafter.
"Rows are non-exclusive" note, verbatim	5×	Same.
"3 linked patents"	16×	Patent Activity (p.42 to 44) says "3 patents, 2 assignees, too small to conclude" five ways across four paragraphs, each with its own Confidence: Low footer.
"118 projects" / "\$125.8M"	20× / 10×	The sample descriptors are restated in nearly every section header paragraph.
Confidence: ... Evidence: ... footers	36×	Right instinct, wrong density. At this rate they stop being read.
The zero-trial finding	6 sections	The report notices this itself on p.42: <i>"The absence is called out explicitly in other sections of this report."</i>

The clearest single example: White Space runs p.11 to p.27, about 16 pages, and its Strategic Implications concludes *"The analyzed sample does not surface strong quantitative gap signals... no category clears the share-ratio and broader-NIH floors."* Sixteen pages to report that nothing cleared the threshold. That is an honest result and it deserves one paragraph.

6 A proposed 10-page core, with everything else on demand

Nothing below is deleted. It is relocated, so the buyer reaches the finding fast and can pull the supporting depth when they want it.

CORE REPORT, TARGET 10 PAGES

- Cover, table of contents, and a **scope-integrity banner** when the pre-flight check fires
- What this does and does not cover (keep, tighten)
- Executive summary, hedges reduced to one line
- Landscape: one table plus funding by year
- The 5 clusters: one table, no per-cluster prose
- White space: **one** table covering all 5 dimensions, no keyword dumps
- Top 10 projects, 3 lines each, **no manufactured relevance line**
- Market context, close to as-is
- Next steps

COMPANION APPENDICES, GENERATED ON REQUEST

- **A.** Methodology, keyword taxonomy, match tiers, all caveat text defined once
- **B.** Full project list with abstracts (all 118)
- **C.** Publications (all 662) and journal distribution
- **D.** Patents and freedom-to-operate
- **E.** Full organization and PI tables (70 and 130)

Each appendix is a selling point in its own right: "full 118-project appendix available" reads as depth. Bundled into the main document it reads as padding.

One structural suggestion. granted.bio is a website, and a 53-page PDF is a strange primary artifact for one. The natural output is a **web report with a sticky table of contents and expandable sections**, where keyword lists and caveats live behind a disclosure triangle, and the PDF is the *export* rather than the product. That also solves the redundancy problem structurally: boilerplate that appears five times in a linear PDF appears once in a tooltip.

7 The question I could not answer: who is this for?

This is where I would most like your input, because it shaped how I read everything above. The cover says **Persona: Researcher**, so personas exist in the product. But the body addresses at least four different readers, in their own voices:

Page	Who the text is talking to
p.4	"For a researcher entering this space, three watchpoints stand out..."
p.4	"For investors and founders scoping pipeline continuity..."
p.27	"For a researcher writing an R01 or R21 in this space..."
p.43	"commercial entrants should not treat the absence of commercial assignees..."

If the persona selector is meant to gate the content, it is not gating it yet. If it is meant to set emphasis only, then the investor and founder language inside a Researcher report is leakage worth hunting.

Where that lands for the use case MJ described

MJ introduced granted.bio to me as the answer to "who needs this?": a lab that just landed an SBIR is a funded, time-boxed, identifiable prospect. That is a **demand-side** pitch. This report is built for the **supply side**. It answers "where should I aim my next grant, and how do I frame it for study section". Those are different products.

The sharpest version of it: p.51 explicitly instructs the reader to use the Key Organizations table "to drive a systematic PubMed and bioRxiv search, not as a contact list." For the use MJ pitched, a contact list is precisely the deliverable. The report is refusing, by design, to do the thing I was introduced to it for.

8 Questions for you

1. **What actually changes between personas?** If I regenerate this same topic as Investor rather than Researcher, does the analysis change, or only the framing paragraphs?
2. **Is there a vendor or service-provider persona?** My question is not "where is the research going", it is "which companies are buying this capability right now". Different output, largely the same underlying data.
3. **Can it surface companies rather than labs?** The buyer I care about is an AI-first protein-design company with a thin internal biology team. Can semantic search over grant and publication data reach those, or does an NIH RePORTER base structurally bias the output toward academic institutions?
4. **What is the intended time-to-value?** At the pricing I have seen quoted, the buyer needs to hit something they could not have googled within about two minutes. Today the single most important finding about this query is on page 17.
5. **How do you plan to catch domain errors at scale?** Every issue in section 3 and 4 is invisible without a subject-matter reader. Cell-free is one narrow vertical; the same class of error presumably exists in the others, and you cannot personally review them all.

9 The short version

Ship the pre-flight scope check first. It costs the least, it prevents the worst outcome (a confident 53-page report on a sample that does not contain the topic), and it turns your biggest failure mode into a trust-building moment.

Then stop generating relevance bridges when there is no link. Saying "no direct link, included for context" costs you less than a wrong sentence and protects the credibility of everything correct in the document.

Then cut to a 10-page core with appendices on demand. The analysis is not too deep. It is that the reference material and the reading material are in the same file, and the reader has to separate them.

The underlying engine clearly works: the cross-linking of projects to publications, patents, and trials is real, the disclosure discipline is better than most commercial research products I read, and the market section is genuinely good. What this particular report exposes is a gap between semantic retrieval and domain truth, on a topic narrow enough to make the gap visible. That is a useful thing for a beta to surface, and it is fixable.

Happy to run two or three more queries against topics I know well and report back the same way, if that is useful to you. Testing where retrieval succeeds is more informative than testing where it fails, and I can tell the difference quickly in this field.

A note on what this review is not. I have deliberately not commented on pricing, packaging, or go-to-market. This is a read of one report by one domain reader, offered as the feedback I promised in exchange for access.